India's Energy Transition

500 GW Renewable Target, Nuclear Expansion, and the Coal Dilemma

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Executive Summary

India's energy transition is the most consequential — and most complex — decarbonisation challenge of any major economy. India is simultaneously the world's third-largest energy consumer (behind China and the United States), the third-largest CO_2 emitter, a country where 800 million citizens are still energy-poor by developed-world standards, and a country that has committed to reaching net-zero emissions by 2070 and 500 GW of renewable energy capacity by 2030 under Prime Minister Modi's COP26 Panchamrit commitments. The central tension: India cannot afford to sacrifice growth for climate, but it also cannot afford to build the carbon infrastructure that its development trajectory would historically demand. Resolving this tension — finding a pathway to full energy access at the living standards India's population deserves without the fossil fuel dependence that comparable historical development required — is the defining policy challenge of India's next quarter-century.

As of August 2026, India has achieved approximately 195 GW of installed renewable energy capacity (solar + wind), making it the world's 4th largest renewable power market. Coal still supplies approximately 70% of India's electricity generation. The gap between the renewable capacity target (500 GW by 2030) and current trajectory (195 GW installed, 45 GW under construction) is significant — India needs to add approximately 260 GW of renewable capacity in four years to meet its target, an installation rate more than double the current pace. The target will likely be missed by 2-3 years but the underlying trajectory — falling solar costs, growing domestic manufacturing, expanding transmission infrastructure — is genuinely positive.

Part I — Solar: The 100 GW Milestone and the 500 GW Target

India's Solar Revolution

India's installed solar photovoltaic capacity crossed 100 GW in January 2024 — a milestone that would have seemed impossible when India's solar mission was launched at 20 GW in 2010. The pace of installation has accelerated dramatically: India added approximately 24 GW of solar in FY2025 alone, driven by utility-scale solar parks (Rajasthan, Gujarat, Andhra Pradesh, Karnataka, Madhya Pradesh), rooftop solar (PM Surya Ghar scheme targeting 10 million households), and industrial captive solar projects.

The economic driver is the collapse of solar module prices. Utility-scale solar power purchase agreement (PPA) tariffs in India have fallen from Rs 7.5/kWh (2011) to Rs 1.99-2.44/kWh in competitive tenders (2026) — making solar the cheapest source of new electricity generation in India. At Rs 2/kWh, solar is competitive with coal-fired power on a levelised cost basis, removing the need for subsidy beyond the upfront capital cost advantage that lower-cost financing provides to coal incumbents.

The Domestic Manufacturing Imperative

India's solar scale creates a strategic vulnerability: approximately 90% of solar modules installed in India through 2022 were manufactured in China. The Approved List of Models and Manufacturers (ALMM) policy and the Basic Customs Duty on solar panels (40%) are India's tools to force domestic manufacturing — requiring government procurement tenders to use only ALMM-listed modules, the vast majority of which are Indian-made. PLI schemes for solar PV modules have committed Rs 4,500 crore to incentivise domestic manufacturing capacity. By 2026, India's domestic module manufacturing capacity has grown to approximately 60 GW per year (still insufficient for 500 GW target without imported equipment) — with Adani Solar, Tata Solar, Waaree Energies, and Vikram Solar as the dominant domestic players.

Part II — Wind Energy and the Offshore Frontier

India's Wind Endowment

India's wind energy resource is among the world's richest: the Indo-Gangetic Plain and western coastal states (Gujarat, Maharashtra) experience reliable winds; Tamil Nadu's southern tip is one of Asia's most productive wind sites; and the Himalayan eastern states have altitude-enhanced wind resources. India's installed wind capacity stands at approximately 47 GW (August 2026), with Tamil Nadu (16 GW+) and Gujarat (10+ GW) the largest wind states.

The challenge: India's best onshore wind sites are increasingly saturated; the next tranche of onshore wind at lower-quality sites is more expensive. Offshore wind — where India has an estimated 70 GW+ of technically recoverable potential in the Gujarat, Maharashtra, and Tamil Nadu coasts — remains essentially undeveloped. The National Offshore Wind Energy Policy (2015) established the framework; the National Institute of Wind Energy has completed wind resource assessments; but the first commercial offshore wind tender has repeatedly been delayed by transmission infrastructure gaps, grid connection costs, and the absence of Indian offshore wind supply chain (installation vessels, offshore substations, marine cables). The target of 30 GW offshore wind by 2030 is effectively unachievable; a more realistic 2030 target would be 5-8 GW.

Part III — Nuclear Energy: The Quiet Renaissance

India's Civil Nuclear Programme

India operates 22 nuclear reactors with a total installed capacity of 7.48 GWe — representing approximately 3% of India's electricity generation. By standards of India's nuclear programme ambition (set out in its three-stage programme devised by Homi Bhabha in the 1950s), this is modest. The atomic energy establishment's vision — PHWR (Pressurised Heavy Water Reactors) to breed plutonium, fast breeder reactors to use plutonium, and ultimately thorium reactors using India's massive thorium reserves — has advanced slowly due to technological complexity, safety standards, cost overruns, and the isolation from international nuclear cooperation that India's status as a non-NPT nuclear weapon state imposed until the 2008 US-India Civil Nuclear Agreement.

The 2026 picture is more positive than the decade-long stagnation of the 2010s suggests. Kudankulam Nuclear Power Plant Units 3-6 (Russian VVER-1000 design) are under construction in Tamil Nadu; Gorakhpur Haryana Anu Vidyut Pariyojana (GHAVP) Units 1-2 are under construction; and a fleet of 10 new 700 MWe PHWRs were approved in 2017 and are at various stages of construction. India's total under-construction nuclear capacity (approximately 8,000 MWe) is among the largest new nuclear construction programmes outside China.

The Small Modular Reactor Opportunity

India's nuclear regulator (AERB) and Department of Atomic Energy are actively evaluating SMR (Small Modular Reactor) technologies — both indigenous designs (Bhabha Atomic Research Centre's AHWR and small PHWR variants) and international collaboration with US (Westinghouse, X-energy), French (EDF), and South Korean (KEPCO) SMR developers. India's 22 operating PHWRs give it deep expertise in modular, standardised reactor construction — making SMR integration technically less challenging than for countries without nuclear operating experience. The potential market: remote industrial sites (aluminium smelters, petrochemical complexes, data centres), replacing diesel-based off-grid generation, and providing baseload support for the intermittent renewable fleet.

Part IV — Coal: The Unavoidable Reality

India's Coal Dependency

India is the world's second-largest coal producer and second-largest coal consumer (after China). Coal Mines India Limited (Coal India) — the world's largest coal mining company, producing approximately 780 million tonnes per year — supplies approximately 55% of India's primary energy. India has approximately 300 billion tonnes of proven coal reserves — sufficient for 200+ years at current production rates. Economically, coal provides the cheapest base-load electricity (Rs 2.5-4/kWh for existing plants with amortised capital costs) and sustains 3-4 million direct mining jobs in some of India's poorest states (Jharkhand, Chhattisgarh, Odisha).

The political economy of coal transition in India is stark: the coal belt states (Jharkhand, Chhattisgarh, Odisha, West Bengal) are among India's poorest; mining employment and royalty revenue are central to state finances; and the affected communities have no alternative employer at equivalent scale. India's COP26 commitment — to phase down coal "consistent with national circumstances" — is honest about the constraint. India will not phase out coal by 2030 or even by 2040. The realistic scenario is that India's coal consumption peaks in the late 2020s and then plateaus — not because of regulatory pressure but because renewables + storage become cheaper at the margin than new coal, while existing coal plants continue operating until end of life.

Part V — Energy Access: The Unfinished Agenda

Electricity for All

India's electricity access story is one of the most dramatic in global development history: in 2015, approximately 300 million Indians had no access to electricity. By 2019, the Saubhagya scheme had achieved near-universal household electrification (99.9% of villages "electrified"). But the gap between a wire reaching a village and reliable 24x7 power supply remains significant — in many rural areas, power is available only 8-12 hours per day, voltage fluctuates, and reliability is poor. Improving power quality — not just access — is the next phase of India's energy equity agenda.

India's energy access expansion has been the world's most consequential driver of electricity demand growth — adding 80-100 TWh of new demand annually from newly connected households and rising per-capita consumption. This demand growth validates India's renewable energy investment (new load requires new generation) while also explaining why India's CO■ emissions continue rising in absolute terms even as the carbon intensity of electricity (CO■ per kWh) falls.

Conclusion: The Trilemma Navigator

India's energy policy must simultaneously navigate three objectives that are partially in tension: energy security (domestic resources, supply chain control), energy equity (affordable, reliable power for all 1.44 billion citizens), and environmental sustainability (decarbonisation commitments). No country has fully solved this trilemma, and India's scale and development stage make it uniquely difficult.

The 2030 picture: India will have approximately 350-400 GW of renewable capacity (missing the 500 GW target but still representing a near-tripling from today), 10+ GW of new nuclear under construction, a coal fleet that has peaked but not declined, and electricity access that is near-universal but improving in quality. The energy transition is real and advancing — but it is an 18-25 year story, not a 5-year one. India's path to net-zero by 2070 requires every subsequent decade to show steeper decarbonisation than the current one. The foundation being built now — in solar manufacturing, nuclear construction, grid infrastructure, and energy storage — determines whether that steeper trajectory is achievable.

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All data from MNRE, CEA, MoP, IEA, Bloomberg NEF, and cited sources.